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# Occupants' indoor environmental quality satisfaction factors as measures of school teachers' well-being

Abdul-Manan Sadick [a, *], Mohamed H. Issa [b]

[a] Faculty of Engineering, Department of Civil Engineering, University of Manitoba, Room E1-368A EITC, 15 Gillson Street, Winnipeg, MB R3T 5V6 Canada
[b] Assistant Professor, Faculty of Engineering, Department of Civil Engineering, University of Manitoba, Room E3-589 EITC, 15 Gillson Street, Winnipeg, MB R3T 5V6, Canada

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ABSTRACT

Most studies that investigated the impact of buildings' indoor environmental quality (IEQ) focused on office buildings, with those studying schools focusing on the impact on students' well-being rather than teachers'. Additionally, most of these studies limited their assessment of well-being to occupants' satisfaction with IEQ factors (e.g. thermal comfort and lighting), overlooking essential aspects related to psychological, social, and physical well-being. This paper presents results of a study conducted in 32 schools in Manitoba, Canada, to investigate the extent to which occupants' IEQ satisfaction factors can be used as measures of teachers psychological, social, and physical well-being. The study involved adapting, deploying and refining an existing IEQ satisfaction survey; and developing, deploying, refining and validating three new surveys to evaluate teachers' psychological, social and physical well-being in schools. Three existing validated well-being surveys in the literature for assessing generic psychological, social and physical well-being were also deployed with the other surveys to teachers in the 32 schools. The association analyses between these different surveys found ventilation and thermal comfort to be the most significant measure of teachers' physical well-being with moderately positive correlations, while lighting and acoustics and privacy were insignificant measures of well-being, thus reinforcing the need to distinguish between occupants' IEQ satisfaction and their well-being in future research. The newly developed well-being surveys include some factors not found in the existing generic well-being surveys and are therefore likely to produce the most representative assessment of teachers' well-being for detailed IEQ impact investigation.



## 1. Introduction

There is compelling evidence in the literature supporting the premise that improving buildings' indoor environmental quality (IEQ) is likely to have a positive impact on occupants' well-being [1–3]. Despite this evidence, most studies [e.g. Refs. [4], [5], [6]] that investigated the impact of IEQ on occupants focused on office buildings. Relatively, fewer studies focused on schools, with most of them [e.g. Refs. [7], [8], [9]] investigating the impact of schools' IEQ on students' performance and well-being rather than teachers' [10]. The few that explored the impact on teachers' well-being [e.g. Refs. [3], [10], [11]] linked improved IEQ to improved teachers' well-being. Another limitation of most existing IEQ studies [e.g. Refs. [12], [13], [14]] is their restriction of the definition of well-being to occupants' IEQ satisfaction, ignoring other essential subdivision of personal well-being such as psychological, social, and physical [15]. There is also little empirical evidence of the extent to which IEQ satisfaction factors such as thermal comfort and lighting assesses occupants' psychological, social and physical well-being, particularly school teachers. Most of the predominant well-being measures found in the literature of the social sciences are not for specific life domains (i.e. context-free) [16] and are thus likely to exclude well-being manifestations that are unique to teachers in school environments.

This study is part of a broader one at the University of Manitoba in collaboration with the Government of Manitoba Public School Finance Board and two Manitoba school divisions with the overall goal of assessing the impact of schools' physical and IEQ conditions on teachers' well-being. The main goal of this study, however, was

* Corresponding author.
*E-mail addresses:* sadicka@myumanitoba.ca (A.-M. Sadick), Mohamed.Issa@umanitoba.ca (M.H. Issa).

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to assess the extent to which existing occupants' IEQ satisfaction measures can be used to assess teachers' psychological, social and physical well-being in school environments. Specific objectives included the following: 1) adapting, deploying, and refining an existing IEQ survey to evaluate teachers' IEQ satisfaction in schools, 2) developing, deploying, refining and validating three new surveys to assess teachers' psychological, social and physical well-being in school respectively, and 3) determining the extent to which the refined IEQ survey assessed teachers' well-being. Refining the IEQ and well-being surveys entailed determining the most significant items required to assess teachers' IEQ satisfaction and well-being respectively in school environments. Validating the new well-being surveys involved deploying three existing validated context-free well-being surveys for psychological, social and physical well-being and investigating the associations between the new and existing surveys. The newly developed and validated teacher's well-being surveys addressed the weakness of the existing context-free well-being surveys given that they include several manifestations of well-being that are unique to teachers in school environments. Determining the extent to which IEQ factors assessed teachers' well-being involved investigating the associations of the refined IEQ satisfaction factors with the newly developed and validated teacher's well-being surveys and the existing validated well-being surveys respectively.

## 2. Background information

This section summarizes the literature review conducted for this study and starts with a definition of IEQ followed by a review of IEQ impact on building occupants. Furthermore, the concept of well-being as it pertains to the IEQ and social sciences literature is discussed in addition to the instruments used to measure well-being.

### *2.1. Indoor environmental quality and its impact on building occupants*

The Center for Disease Control and Prevention (CDC) [17] defined IEQ as "the quality of a building's environment in relation to the health and well-being of those who occupy space within it". Most studies that investigated the impact of IEQ on occupants focused on office buildings and in line with the CDC definition, some of them [e.g. Refs. [18], [19]] investigated the impact on occupants' health. For example, Witterseh, et al. [18] found that several sick building syndrome symptoms including headache, throat and eye irritation worsened with increasing indoor air temperature. Several studies [e.g. Refs. [20], [21], [22]] also investigated the impact of office buildings' IEQ on employee's performance. A desktop study [22] simulated office tasks (such as reading, typing, and addition) and predicted that if indoor air quality was improved such that only 10% of office workers were dissatisfied, the annual increase in productivity would be 10 times that of the required increase in energy and maintenance costs. de Korte et al. [5] studied the impact of pre-set radiant heat on office tasks using the dual visual memory task consisting of a reading task followed by a computer task and found no significant impact on the performance of those tasks. In a more recent study, Maula, et al. [6] investigated the impact of slightly warmer temperature (i.e. 29 °C) on work performance in a mock open-plan office and found a negative effect on working memory. These studies relied on measures of cognitive performance as a proxy to assess the performance of office workers due to the difficulty of measuring that performance in real time. Others [e.g. Refs. [23], [24]] investigated the impact of IEQ on the self-assessed productivity of office workers, raising concerns about the objectivity of that assessment as self-assessed productivity is likely to be exaggerated.

Most of the IEQ studies in schools [e.g. Refs. [8], [10], [25]] focused on students rather than teachers [10]. Issa et al. [8] noticed decreased absenteeism and increased student performance in green schools compared to conventional and energy-retrofitted schools. Both Sörqvist [26] and Boman [27] found a negative effect on student learning resulting from intelligible speech. Tanner [28] showed a positive relationship between daylighting and students' academic achievement. For teachers, Madureira, et al. [29] found a significant association between the concentration of indoor air pollutants (e.g. carbon dioxide and total volatile organic compound) and the prevalence of fatigue, headaches, and thinking difficulties. More recently, Kielb, et al. [10] found a significant association between the prevalence of asthma amongst teachers and indoor air quality problems such as mold and dust.

### *2.2. Well-being in the IEQ literature*

In the IEQ literature, the concept of occupant well-being is not explicitly defined. Occupant well-being, often termed occupant comfort or satisfaction usually refers to satisfaction with the four main IEQ factors namely indoor air quality, thermal comfort, acoustics, and lighting [12–14,30] and is generally assessed using questionnaire surveys. Several IEQ surveys have been developed including the Center for the Built Environment (CBE) IEQ survey [31], the Health Optimisation Protocol for Energy-efficient Buildings (HOPE) survey [32], the Building Use Studies (BUS) survey [33], the Building In Use (BIU) survey [34], Building Occupants Survey System Australia (BOSSA) [35], and the National Research Council Canada (NRC) environmental feature ratings [36]. The construction of questions for specific IEQ factors in the different surveys varies, however, they are not conceptually divergent. For example, the CBE survey include the thermal comfort question "How satisfied are you with temperature in your workspace?" [37, p. 393], while the HOPE and BOSSA surveys respectively include "How would you describe typical working conditions in the office … Temperature in winter" [32, p. 4], and "Please rate your satisfaction with the temperature conditions of your normal work area … ?" [35, p. 205].

Bluyssen et al. [12] studied IEQ satisfaction using 5732 office building respondents from the HOPE survey database and concluded that occupants' well-being cannot be limited to their satisfaction with IEQ factors. Additionally, the environmental comfort model proposed by Vischer [38] reached a similar conclusion, with the model defining occupants' comfort as a function of physical, functional, and psychological comfort and not just occupants' satisfaction with IEQ factors. Some of the IEQ surveys include items that assess occupants' personal well-being beyond IEQ satisfaction. The BOSSA survey has an item for assessing the impact of work environment on occupants general health [35]. The BUS survey also includes three items respectively for assessing general health, overall well-being, and stress levels with respect to indoor work environments [33]. However, these well-being assessments are less detailed and will provide a limited assessment of occupants' well-being. The HOPE survey includes a whole section on personal well-being, however, the items are focused mainly on sick building syndromes and their frequencies [32]. Only a few IEQ studies [e.g. Refs. [4], [39], [40]] have explored occupants well-being using surveys from the social sciences; however, these studies focused on offices. Newsham et al. [4] found that respondents of green office buildings experienced more positive feelings relative to those of conventional office buildings. Additionally, Schreuder, et al. [40] found a significant positive change in the well-being of respondents who moved from old office buildings to new ones with superior IEQ. Only Aries et al. [39] investigated the relationship between occupants' IEQ satisfaction and their well-being using surveys from the social sciences and found a significant association between view quality and physical and psychological comfort. These results reinforce the need for research focused on how IEQ satisfaction relates to or assesses occupants' psychological, social, and physical well-being to address the limited empirical evidence on this topic, particularly for school teachers.

### 2.3. Well-being in the social science literature

The World Health Organization (WHO) defines health as "a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity" [41]. Dodge et al. [15] saw individual well-being as "the balance point between an individual's resource pool (i.e. psychological, social, and physical) and the challenges faced (i.e. psychological, social, and physical)". For building occupants, that balance can be disturbed by adverse indoor environmental conditions [39,42]. Review of the predominant well-being measures in the social sciences confirms Bluyssen et al. [12] assertion that occupants' well-being cannot be limited to IEQ satisfaction. For example, Ryff's Scale of Psychological Well-Being includes the factors "positive relations with others" and "environmental mastery" [43, p. 1072]. Additionally, Keyes' Five Factors Social Well-being Model includes the factors "social integration" and "social contribution" [44, pp. 138,139]. Furthermore, the RAND-36 survey's physical well-being segment includes the factors "physical functioning" and "energy/fatigue" [45, P. 357]. These factors of personal well-being are essential to the well-being of all building occupants, including school teachers, thus calling into question the extent to which IEQ satisfaction factors assess occupants' well-being. Most of the well-being measures found in the social sciences are composed of generic assessment items (i.e. context-free), thus, making them applicable to different life domains such as home, school, and offices. However, the complex social web in school where teachers are engaged in nurturing the academic and social development of students in collaboration with their colleagues and superiors is distinct from the context of homes and offices. Hence, the context-free well-being measures in literature will include items relevant to teachers in school, but will also exclude items unique to school environments and as such may not adequately capture teacher's well-being.

## 3. Methods

This research was conducted in two school divisions in the province of Manitoba (Canada) with a total of 32 schools and 852 teachers. The schools were composed of new and old schools with start dates of operation ranging from 1950 to 2012. Most of the schools were constructed with concrete block walls while others were a mix of concrete block walls and drywall partitions. The schools were all mechanically ventilated, however, some of the old schools had no operable windows. Table 1 presents average values of IEQ physical measurement data from 58 classrooms in a sample of 11 schools. The table presents a general view of the IEQ conditions of classrooms in most of the 32 schools.

The research adopted an exploratory sequential mixed-method design that started with qualitative data collection and analysis (i.e. stage one) and followed by quantitative data collection and analysis (i.e. stage two) [46] as shown in Fig. 1.

### 3.1. Stage one

The qualitative stage used semi-structured interviews to identify the manifestations of teachers' psychological, social, and physical well-being in school environments and subsequently used to develop the new preliminary well-being surveys. Invitation emails were sent to all the teachers in the 32 schools to voluntarily participate in these interviews. Of the 25 positive responses, 20 were selected based on school grade levels, years of teaching experience, gender, and age to attain sample variation. The 20 interview participants were from 10 of the 32 schools.

#### 3.1.1. Semi-structured interviews

Semi-structured interviews were conducted following the example of Dagenais-Desmarais and Savoie [16]. The questions were divided into three sections as follows: psychological well-being, social well-being, and physical well-being. For each section, predefined open-ended questions based on literature [47] were used to identify the relevant manifestations of personal well-being, while follow-up questions explored the manifestations most likely to influence the work of teachers in their classrooms. The predefined questions were approved by the University of Manitoba Research Ethics Board. Each interview was audio recorded and ranged from 30 min to one hour.

#### 3.1.2. Analysis of interview data

Following examples [16,47] in literature, the interview data were analyzed using a mix of content analysis and grounded theory analytical strategies. Data fragments (base codes) representing manifestations of well-being were extracted and assigned descriptive labels related to the well-being subdivision of interest. About 75% of the base codes emerged from the first 10 interviews. The base codes for psychological, social, and physical well-being were respectively integrated to form categories. Constant comparative analysis was conducted within and between categories to identify similarities and differences for the creation of subcategories. The subcategories represented the different dimensions of well-being manifestations for the three subdivisions of personal well-being. The number of well-being manifestations included in the preliminary well-being surveys was based on a number of recommendations [47] including the number of respondents who reported a specific manifestation and the ability of a manifestation to capture the meaning of other manifestations. Non-school related well-being manifestations were omitted. The preliminary teachers' psychological (TPWB), social (TSWB), and physical well-being (TPhWB) surveys included 34, 32, and 24 assessment items respectively. The interview transcripts were analyzed using the QSR Nvivo version 10 software.

### 3.2. Stage two

Stage two involved administering an online survey composed of the three preliminary well-being surveys (survey A), the three existing well-being surveys (survey B), and the adapted IEQ satisfaction survey (survey C). All three surveys were administered together and only once to all teachers in the two participating school divisions and were divided into four modules as shown in Table 2. The survey was administered using the Qualtrics online platform. To minimize the survey completion time, each respondent was assigned module 1 and an additional module randomly

**Table 1**
Average values of selected IEQ parameters for a sample of 11 schools.

| IEQ parameters | Mean | Standard deviation |
|---|---|---|
| Carbon dioxide | 1089.19 | 251.65 |
| Relative humidity | 26.35 | 6.62 |
| Air temperature | 22.08 | 0.99 |
| Background noise dB (A) | 28.22 | 3.85 |
| Reverberation time (RT60 at 1 KHz) | 0.53 | 0.18 |
| Daylight factor | 1.16 | 1.01 |

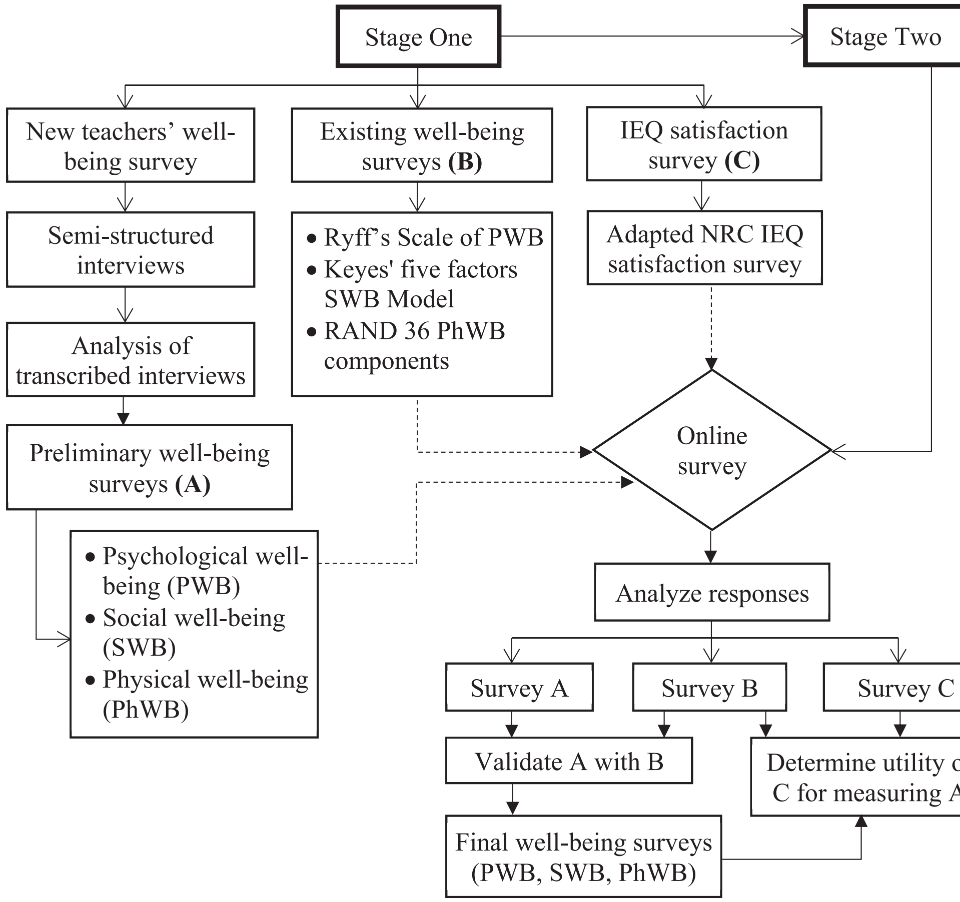


**Fig. 1.** Outline of two stage research methodology.

assigned using the Qualtrics module randomizer set to even distribution. The randomization strategy employed in this research is consistent with other studies [e.g. Ref. [4]] that administered lengthy online surveys. A total of 234 responses were received, representing a response rate of 27.46%, and each subject was surveyed only once. All 234 respondents were assigned module 1, while 80, 78, and 76 were assigned modules two, three, and four respectively. The survey was administered during winter of 2016.

*3.2.1. Occupant IEQ satisfaction survey*

The IEQ satisfaction survey used in this study was adapted from the NRC's environmental features rating originally developed for office occupants. The wording of the 18 items in the NRC survey was modified to suit the context of schools for teachers. The item "Amount of lighting on the desktop" [48, p. 20] was replaced with two items: "Amount of daylight in your classroom" and "Amount of electric light in your classroom"' thus increasing the number of items by one. The preamble to the 19 items was "For each of the following, please select the appropriate checkbox that best expresses your level of satisfaction with the …". The assessment was based on a seven-point scale from "Very Unsatisfactory" (−3) to "Very Satisfactory" (+3) adopted from the NRC survey.

*3.2.2. New preliminary well-being surveys*

Each of the three preliminary well-being surveys assessed one subdivision of teachers' well-being as shown in Table 2. The preceding preamble to the assessment items of each subdivision of well-being was: "For each statement below, indicate which answer best describes your present agreement or disagreement with each statement" [49, p. 90]. In line with the predominant well-being measures in literature, these items were assessed on a six-point scale from "Strongly Disagree" (1) to "Strongly Agree" (6).

*3.2.3. Existing well-being validation surveys*

Three existing validated well-being surveys in the literature were used to validate the newly developed TPWB, TSWB, and TPhWB surveys. These included Ryff's scale of psychological well-being [49, p. 90], Keyes' five factors social well-being model [44, pp. 138,139], and RAND 36 physical well-being components [45, P. 357, 50, p. 443].

*3.2.4. Data analysis*

The objectives of the analysis were as follows: 1) to refine the adapted IEQ satisfaction and the three preliminary well-being surveys for assessing teachers' IEQ satisfaction and well-being respectively, 2) to assess the validity and reliability of the three final well-being surveys, and 3) to assess the extent to which the refined IEQ satisfaction factors assesses teachers' well-being in school environments.

Litte's MCAR test was used to assess the distribution of missing data for each component of the online survey. The threshold for excluding cases with missing data was five percent and above, after which the expectation maximization (EM) algorithm was used to impute missing data [51]. The EM algorithm was preferred given that its estimation is based on the variability inherent in the existing data [51]. Univariate outliers were identified using standardized values with an absolute of 3.29 as cut-off [51]. The Mahalanobis distance statistics at $p < 0.001$ was used to identify

**Table 2**
Online survey components.

| Modules | Name | Content |
|---|---|---|
| Module 1 | Environmental Feature Rating | NRC's environmental features rating (adapted) [36]<br>Demographic information |
| Module 2 | Psychological Well-being | Preliminary TPWB survey<br>Ryff 's scale of psychological well-being [43] |
| Module 3 | Social Well-being | Preliminary TSWB survey<br>Keyes' five factor social well-being model [44] |
| Module 4 | Physical Well-being | Preliminary TPhWB survey<br>RAND 36 physical well-being components [45] |

multivariate outliers and excluded from the analysis [51]. Absolute values above 3 and 8 for skewness and Kurtosis respectively were the thresholds for univariate normality. Multivariate normality was examined using a scatter plot of chi-square versus Mahalanobis distance [52].

As part of refining the adapted IEQ survey, a confirmatory factor analysis (CFA) was conducted to assess the model fit between the NRC survey's three-factor structure of IEQ satisfaction [36] and the teachers' IEQ data in this study. The objective of the CFA was to determine the suitability of the NRC's three-factor structure of IEQ satisfaction (developed for office occupants) for assessing teacher's IEQ satisfaction in schools. Following the established CFA convention [36], multiple indices were used to determine the model goodness of fit. The results showed a lack of model fit; hence, an exploratory factor analysis (EFA) was conducted to determine a suitable factor structure for teachers IEQ satisfaction. The EFA conducted for the IEQ data followed the same procedures described below for the well-being surveys.

The inductive process employed in developing the preliminary well-being surveys was to ensure that the final surveys will be grounded in teachers' conception of well-being in schools. As part of refining the preliminary surveys, EFAs were conducted to identify the significant items and factor structures of the final TPWB, TSWB, and TPhWB respectively, using the 34, 30, and 24 items of their respective preliminary well-being survey. Hence, the objective of the EFAs was to make the final well-being surveys parsimonious by excluding items of the preliminary surveys that were conceptually insignificant in measuring teacher's well-being in school. Preliminary data screening for each EFA was conducted using a correlation matrix and the Kaiser-Meyer-Olkin's (KMO) measure of sampling adequacy for individual items. The inclusion criteria for variables were a correlation coefficient greater than 0.3 with multiple items and a KMO threshold of 0.5. The number of factors to extract was determined using multiple methods including eigenvalue greater than one criterion, scree plot, and parallel analysis. Based on the range of suggested number of factors, multiple EFAs were conducted to determine the preferred solution. Considering the relatively small sample size in this research, principal axis factoring was the preferred extraction method given that it is less sensitive to minimal violations of multivariate normality [53]. Additionally, the orthogonal rotation promax was preferred given that well-being factors were expected to correlate. The preferred solutions were factor structures with the least cross loadings, minimum of three items per factor, and a minimum item loading of 0.4 [53]. Items were also excluded for parsimony considerations and conceptual conflict with assigned factor labels [16].

A two-tailed correlation test was used to assess the convergent validity of the final TPWB, TSWB, and TPhWB survey respectively with its corresponding existing survey. This was to determine the suitability of the newly developed surveys as measures of teachers' well-being. A two-tailed test was preferred to concurrently determine the likelihood of either a convergent (significant positive) or divergent (significant negative) validity between corresponding well-being surveys. The internal consistency and reliability of the final well-being surveys were determined using Cronbach's alpha ($\alpha$). The $\alpha$ values were calculated to determine the likelihood of attaining consistent results when the final well-being surveys are deployed in future. Finally, the associations of the refined IEQ satisfaction factors with the existing well-being surveys and the final teachers' well-being surveys were assessed using two-tailed correlation tests. The objective of these correlation tests was to assess the extent to which the refined IEQ satisfaction factors assesses teachers' well-being.

Individual IEQ satisfaction factors and well-being scores were calculated as an average of the retained items for each of the final factor solutions after reverse coding of negatively keyed items (i.e. items with a negative sign in parenthesis). The statistical analyses were conducted using IBM SPSS version 23, however, the parallel analyses were conducted using O'Connor's [54] SPSS program.

## 4. Results

This section presents results of the analysis conducted to refine the adapted IEQ survey and the preliminary teachers' well-being surveys, as well as the convergent validity of the final well-being surveys and the suitability of the IEQ factors as measures of occupants' well-being.

### 4.1. *Factor structure of teachers' IEQ satisfaction survey*

This subsection presents results of both the CFA and EFA conducted for the adapted IEQ satisfaction survey data. A total of 189 cases were considered for the analysis after excluding cases with more than five percent missing values, univariate outliers, and multivariate outliers.

#### 4.1.1. *Exploratory factor analysis*

Confirmatory factor analysis (CFA) showed a lack of model fit between the NRC's three-factor solution of IEQ satisfaction [36] and the teachers' IEQ satisfaction data. The chi square value of 533.753 was statistically significant, thus showing a significant divergence between the three-factor solution and the current data. Moreover, the normed fit index, the comparative fit index, and the root mean square error of approximation were 0.72 (less than 0.9), 0.77 (less than 0.95), and 0.12 (greater than 0.06) respectively. The model misfit is attributable to the fact that the NRC's three-factor structure was developed with responses from office building occupants. Hence, an EFA was conducted for the current data to identify the most significant IEQ factors and items that are relevant to school teachers. Examination of the correlation matrix showed the absence of singularity and multicollinearity, while the KMO measure of sampling adequacy for individual items were all above the 0.5 benchmark. The omnibus KMO measure of sampling adequacy of 0.883 exceeded the 0.6 cut-off value. The preliminary EFA for the

19 items with principal axis factoring suggested extracting five factors based on the eigenvalue greater one rule. Similarly, the scree plot suggested extraction of five factors. However, parallel analysis for 1000 permutations of the raw teachers' IEQ data suggested extraction of two factors. Hence, multiple EFAs were conducted in turn using principal axis factoring for extraction and altering the number of factors to extract from two to five. The preferred factor structure was a three-factor solution with 12 items accounting for 65.63% of the total variance as shown in Table 3.

The inter-factor correlation coefficients exceeded the threshold of 0.32 [51], thus supporting the use of the Promax orthogonal rotation. All the item loadings (see Table 3) were greater than or equal to 0.4, except the item "Amount of background noise (i.e. not speech) you hear in your classroom" which had a loading of 0.35 under factor three: "Acoustics and Privacy". This item was exceptionally included, considering that it is one of the fundamental variables of building acoustics assessment. Each of the factors in Table 3 accounts for a significant proportion of total variance and have significant α values that attest to their satisfactory internal consistency and reliability. The labels for the three factors: "Lighting" ($L_{sat}$), "Ventilation and Thermal Comfort" ($VTC_{sat}$), and "Acoustics and Privacy" ($AP_{sat}$) were adopted from the NRC's [36] three-factor solution given that the items loading to the respective factors in this study suggested no conceptual change.

### 4.2. Factor structures of teachers' well-being surveys

This subsection presents results of the EFAs conducted to refine the preliminary teachers' well-being surveys. The number of cases considered for the analysis after excluding cases with more than five percent missing values, univariate outliers, and multivariate outliers was 61, 63, and 54 for TPWB, TSWB, and TPhWB respectively. Additionally, two variables were excluded from the analysis of TSWB due to the lack of univariate normality.

#### 4.2.1. Exploratory factor analysis

Review of the correlation matrices and the KMO measure of sampling adequacy for individual items reduced the number of variables from 34 to 26, 30 to 27, and 24 to 20 for TPWB, TSWB, and TPhWB respectively. The omnibus KMO measure of sampling adequacy for the remaining variables were 0.74, 0.80, and 0.70 respectively for TPWB, TSWB, and TPhWB, thus, exceeding the threshold value of 0.6 [51]. Results of the EFAs that followed are reported below.

*4.2.1.1. Teachers' psychological well-being.* The Initial EFA of the 26 variables of TPWB suggested extraction of eight factors based on the eigenvalue greater than one rule. Additionally, the scree plot suggested extraction of four factors whereas parallel analysis suggested extraction of two factors. Consequently, multiple EFAs were conducted by varying the number of factors to extract from two to eight. The preferred solution for TPWB was a two-factor solution with 12 items, accounting for 50.25% of the total variance observed as shown in Table 4. The inter-factor correlation of 0.51 exceeded the recommended threshold of 0.32 for correlated factors, thus supporting the choice of the promax rotation method. All the item loadings (Table 4) exceeded the threshold of 0.4, thus contributing significantly to the conceptual meaning of their respective latent factors. Additionally, the α values affirm the internal consistency of both factors, and each of them accounted for a significant proportion of the total variance. These two factors were labelled as F1 – "Job satisfaction and professionalism" and F2 – "Felling of accomplishment and recognition".

*4.2.1.2. Teachers' social well-being.* The eigenvalue greater than one rule and the scree plot suggested extraction of seven and eight factors respectively from the initial 27 variables of TSWB, while parallel analysis suggested three factors. Following multiple EFAs, the final solution consisted of three factors with 12 items and accounting for 74.10% of total variance as shown in Table 5. The three factors are F1 – "Social Functioning in School", F2 – "Social Connectedness in School", and F3 – "Classroom Socialization". Additionally, each factor accounted for a significant proportion of total variance and had satisfactory internal consistency as evidenced by the α values, while factor loadings were all above 0.4. Although the inter-factor correlation of 0.23 between F1 and F3 was marginally below the 0.32 threshold, F3 significantly correlated with F2, validating thereby the choice of the promax rotation.

*4.2.1.3. Teachers' physical well-being.* For TPhWB, the initial number of factors to extract based on eigenvalue greater than one, scree plot, and parallel analysis were six, nine, and two respectively. The preferred result after multiple EFAs consisted of two factors with 11 items and accounting for 56.23% of total variance as shown in Table 6. These two factors are F1 – "Role and Work Environment Outcomes" and F2 – "Physical Functioning in Class". The inter-factor correlation of 0.50 confirms the choice of the promax rotation. The two factors have satisfactory item loadings, significant internal consistency values, and each account for a significant proportion of total variance (see Table 6).

### 4.3. Validity and reliability of the final teacher's well-being surveys

Table 7 presents the two-tailed Pearson correlation coefficients between the final TPWB, TSWB, TPhWB surveys and their corresponding existing well-being surveys. The coefficients of correlation were all moderately positive and statistically significant. The levels of association with existing well-being measures found in this study are within the range of values (i.e. $r = 0.19$ to $0.53$) reported in the literature for similar convergent validity studies [16]. Hence, each of the three measures of teachers' well-being had satisfactory convergent validity with its corresponding existing measure. The α values for the final well-being surveys were 0.87, 0.90, and 0.80 for TPWB, TSWB, and TPhWB respectively, thus, exceeding the threshold of 0.70 for an acceptable level of internal consistency and reliability [51].

### 4.4. Refined IEQ satisfaction factors as measures of teacher's well-being

Table 8 presents results of the tests of associations of the refined IEQ satisfaction factors with the existing well-being surveys and the final teachers' well-being surveys. Apart from $VTC_{sat}$ which significantly correlated with the existing physical well-being measure (i.e. RAND 36), $L_{sat}$ and $AP_{sat}$ did not significantly correlate with any of the existing well-being measures. Additionally, $VTC_{sat}$ correlated significantly with the three final teachers' well-being measures, while $L_{sat}$ and $AP_{sat}$ significantly correlated with TPhWB only.

## 5. Discussions

The refined IEQ satisfaction survey developed in this study was compared to the original NRC IEQ survey developed by Veitch et al. [36]. Although the difference in sample size makes direct comparison insignificant, the CFA and EFA results in this study seem to project the uniqueness of teachers' perception of IEQ satisfaction. The factor labels in this study were adopted from Veitch et al. [36], however, the item compositions and factor loadings varied significantly. The $L_{sat}$ factor in this study includes the item "Amount of daylight in your classroom" that had the highest factor loading,

**Table 3**
12 items three-factor solution for teachers' IEQ satisfaction survey.

| Item | Factor | | | Item variance |
|---|---|---|---|---|
| | F1 | F2 | F3 | |
| *F1 — Lighting* | | | | |
| 1-Amount of daylight in your classroom | 0.93 | | | 0.74 |
| 15-Your access to a view of outside from your classroom | 0.77 | | | 0.59 |
| 17-Quality of lighting (both electric and daylight) in your classroom | 0.72 | | | 0.61 |
| 2-Amount of electric light in your classroom | 0.40 | | | 0.31 |
| *F2 — Ventilation and Thermal Comfort* | | | | |
| 4-Temperature in the classroom where you teach regularly | | 0.80 | | 0.55 |
| 13-Air movement in your classroom | | 0.73 | | 0.66 |
| 14-Your ability to alter or control physical conditions (such as temperature, light etc.) in your classroom | | 0.72 | | 0.47 |
| 3-Overall air quality in your classroom (i.e. smells bad, stuffy, stale air etc.) | | 0.67 | | 0.54 |
| *F3 — Acoustics and Privacy* | | | | |
| 7-Level of visual privacy within your classroom | | | 0.99 | 0.74 |
| 6-Level of privacy for both teaching and non-teaching related conversations in your classroom | | | 0.89 | 0.71 |
| 19-Degree of enclosure of your work area by walls, screens or furniture | | | 0.55 | 0.50 |
| 10-Amount of background noise (i.e. not speech) you hear in your classroom | | | 0.35 | 0.28 |
| Percentage of variance explained (%) | 40.30 | 14.77 | 10.56 | |
| Cronbach's alpha ($\alpha$) | 0.81 | 0.82 | 0.79 | |
| Inter-factor correlations | | | | |
| F1 | 1 | | | |
| F2 | 0.50 | 1 | | |
| F3 | 0.35 | 0.56 | 1 | |

**Table 4**
12 Items two-factor solution for the new final TPWB survey.

| Item | Factor | | Item variance |
|---|---|---|---|
| | F1 | F2 | |
| *F1 — Job satisfaction and professionalism* | | | |
| 25-My classroom is not fit for teaching and learning (−) | 0.75 | | 0.43 |
| 3-I am able to cope with everyday student interactions in class | 0.67 | | 0.37 |
| 19-I know my responsibilities as a teacher in school | 0.67 | | 0.54 |
| 14-I get all the support I need to do my work in school | 0.67 | | 0.65 |
| 12-I feel secured to perform my duties as a teacher | 0.59 | | 0.48 |
| 30-My workload in school is too much (−) | 0.51 | | 0.31 |
| *F2 — Feeling of accomplishment and recognition* | | | |
| 18-I help my students to get the support they need | | 0.67 | 0.44 |
| 11-I feel eager to make a difference in my school | | 0.67 | 0.43 |
| 12-I like the occasional compliments and feedback from my colleagues | | 0.61 | 0.33 |
| 28-My students appreciate my support | | 0.57 | 0.31 |
| 33-The achievements of my students are gratifying | | 0.55 | 0.30 |
| 29-My students' personal problems are mine too | | 0.54 | 0.30 |
| Percentage of variance explained (%) | 35.07 | 15.18 | |
| Cronbach's alpha ($\alpha$) | 0.81 | 0.75 | |
| Inter-factor correlations | | | |
| F1 | 1 | | |
| F2 | 0.51 | 1 | |

thus, making it the most significant lighting item ahead of both access to a view of outside and quality of lighting. This outcome agrees with earlier IEQ studies in schools [e.g. Ref. [28]] that found a significant association between students' performance and daylit classrooms, thus, highlighting the significance of the amount of daylight to teachers' satisfaction with the quality of lighting. In contrast, Veitch, et al. [36] had no separate item for daylight, while the quality of lighting and access to a view of outside were the most and least significant items respectively.

Although the refined teachers' IEQ satisfaction survey developed in this study contextualizes teachers' IEQ satisfaction to school environments, the survey is conceptually in line with the IEQ literature. The four items of the $VTC_{sat}$ factor in this study includes all three items of Veitch et al.'s [36] $VTC_{sat}$ factors in addition to the item “Your ability to alter or control physical conditions (such as temperature, light etc.) in your classroom”. This item had a significant factor loading of 0.72 under the $VTC_{sat}$ factor. Although the item included the control of other indoor space parameters such as the amount of light, the highly significant loading under the $VTC_{sat}$ factor suggests that the teachers were probably more concerned with controlling indoor air temperature. This result is not surprising considering that most classrooms in this study had no thermostat for decentralized control of temperature, while a significant number of the classroom had non-operable windows, and is in line with the findings of others [e.g. Refs. [55], [56]] in the IEQ literature. Visual privacy and speech privacy were the most significant items of the $AP_{sat}$ factor with factor loadings of 0.99 and 0.89 respectively, demonstrating their relative importance to teachers who seem to prefer an indoor environment that minimizes distractions from outside, and from nearby classrooms. The significance of speech privacy in this study resonates with the results of both Sörqvist [26] and Boman [27] who found a negative association between speech intelligibility and student performance.

The results also confirmed that the final TPWB, TSWB, and

**Table 5**
12 items three-factor solution for the new final TSWB survey.

| Item | Factors | | | Item variance |
|---|---|---|---|---|
| | F1 | F2 | F3 | |
| *F1 — Social Functioning in School* | | | | |
| 30-We figure out a way to work through our differences | 0.91 | | | 0.71 |
| 29-There is mutual respect amongst people in my school | 0.83 | | | 0.74 |
| 19-In my school, learning is a collective responsibility | 0.79 | | | 0.60 |
| 22-In my school, you see people caring about other people | 0.72 | | | 0.61 |
| 31-We get to interact and laugh together as colleagues | 0.64 | | | 0.56 |
| *F2 — Social Connectedness in School* | | | | |
| 8-I feel connected to my colleagues in school | | 0.96 | | 0.87 |
| 14-I have positive interactions with people in my school | | 0.92 | | 0.81 |
| 7-I feel comfortable expressing my feelings to some people in school | | 0.69 | | 0.73 |
| 5-I enjoy the company of people in my school | | 0.68 | | 0.66 |
| *F3 — Classroom Socialization* | | | | |
| 9-I get close and have conversation with students | | | 0.83 | 0.65 |
| 4-I contribute to the social development of my students | | | 0.79 | 0.62 |
| 17-I purposely plan for classroom social interactions | | | 0.46 | 0.39 |
| Percentage of variance explained (%) | 47.23 | 18.15 | 8.56 | |
| Cronbach's alpha ($\alpha$) | 0.89 | 0.90 | 0.72 | |
| Inter-factor correlations | | | | |
| F1 | 1 | | | |
| F2 | 0.56 | 1 | | |
| F3 | 0.23 | 0.55 | 1 | |

**Table 6**
11 items two-factor solution for the new final TPhWB survey.

| Item | Factor | | Item variance |
|---|---|---|---|
| | F1 | F2 | |
| *F1 — Role and Work Environment Outcomes* | | | |
| 11-I get allergic reactions in school due to conditions of my work environment (−) | 0.98 | | 0.76 |
| 9-I frequently get running nose, dry throat and lips, or watery eyes in school (−) | 0.81 | | 0.68 |
| 14-I have bodily pains that affect my work in school (−) | 0.60 | | 0.32 |
| 1-I always feel tired in class (−) | 0.58 | | 0.48 |
| 12-I get bodily pains from standing and movements in class (−) | 0.57 | | 0.56 |
| 8-I feel uncomfortable due to the level of air temperatures in my classroom (−) | 0.54 | | 0.30 |
| 10-I frequently raise my voice in class due to high background noise (−) | 0.46 | | 0.30 |
| *F2 — Physical Functioning in Class* | | | |
| 4-I engage in fun activities with my students in school | | 0.70 | 0.38 |
| 20-I wish I could do less walking in the classroom (−) | | 0.70 | 0.51 |
| 5-I feel alert and energetic in my classroom | | 0.58 | 0.54 |
| 6-I feel comfortable standing, bending, kneeling or sitting on the floor to help my students | | 0.53 | 0.35 |
| Percentage of variance explained (%) | 41.48 | 14.75 | |
| Cronbach's alpha ($\alpha$) | 0.85 | 0.73 | |
| Inter-factor correlations | | | |
| F1 | 1 | | |
| F2 | 0.50 | 1 | |

**Table 7**
Validity of teachers' final well-being surveys.

| Validation Measures | Correlation coefficients and significance level | | |
|---|---|---|---|
| | TPWB | TSWB | TPhWB |
| Ryff's scale of psychological well-being | 0.40** (0.001) | | |
| Keyes' social well-being model | | 0.30* (0.018) | |
| RAND 36 physical well-being components | | | 0.47** (0.000) |

Note: ** denotes significant correlation at level 0.01; * denotes significant correlation at level 0.05; Values in parenthesis denotes level of significance for correlation coefficients.

TPhWB surveys developed as part of this research are valid measures of the subdivision of teacher's well-being for which they were intended. The surveys demonstrated satisfactory convergent validity with their corresponding established well-being surveys in the literature. The $\alpha$ values of the three new final well-being surveys attest to their satisfactory internal consistency and reliability. The moderately positive correlations between the newly developed teacher's well-being surveys and their corresponding existing surveys affirm the uniqueness of the new teacher's well-being surveys. Very strong positive correlations would have implied that the new well-being surveys are not significantly different from the existing context-free well-being surveys. For example, the TSWB survey includes the factor “Social Connectedness in School” which is directly comparable to the “Social Integration” factor in Keyes'

**Table 8**
Associations of refined IEQ satisfaction factors with the existing well-being surveys and the final teacher's well-being surveys.

| Existing and new well-being measures | IEQ satisfaction factors | | |
|---|---|---|---|
| | $L_{sat}$ | $VTC_{sat}$ | $AP_{sat}$ |
| Ryff's scale of psychological well-being | 0.004 (0.977) | 0.081 (0.836) | 0.027 (0.836) |
| Keyes' social well-being model | 0.000 (0.360) | −0.177 (0.360) | −0.228 (0.073) |
| RAND 36 physical well-being components | 0.147 (0.288) | 0.341* (0.012) | 0.090 (0.519) |
| Final TPWB | 0.213 (0.099) | 0.280* (0.029) | 0.200 (0.121) |
| Final TSWB | 0.124 (0.333) | 0.347** (0.005) | 0.231 (0.069) |
| Final TPhWB | 0.359** (0.008) | 0.612** (0.000) | 0.376** (0.005) |

Note: ** denotes significant correlation at level 0.01; * denotes significant correlation at level 0.05; Values in parenthesis denotes level of significance for correlation coefficients above.

social well-being model. However, the TSWB survey also includes the factor "Classroom Socialization" which is conceptually related to the Keyes' social well-being model in general but is unique to teachers' social well-being in school. Hence, the moderately positive correlations confirm the difference between final teacher's well-being surveys and their corresponding existing surveys.

The preliminary well-being surveys used for developing the final well-being surveys were established from interviews of 20 out of the 852 teachers. Hence, the preliminary well-being surveys were probably not adequately representative of teachers' well-being manifestations in the population. However, given that the first 10 interviews produced about 75% of the preliminary well-being surveys' base codes, it seems fair to assume that the sample of 20 was reasonably saturated. Hence, additional interviews could have led to new base codes and probably different composition of items in the final well-being surveys, but plausibly including most of the current items. The preliminary well-being surveys were administered to the 32 schools including the 10 schools that participated in the interviews, thus, questioning the validity and generality of the final well-being surveys beyond the sample of schools in this study. Due to restrictions by the University of Manitoba Ethics Review Board, it was not possible to identify and exclude the 20 teachers who participated in the interview from completing the online survey or excluded their responses if they completed the survey. However, approximately 71% of the responses included in the analysis were from the 23 schools that did not participate in the interviews and would likely minimize the concerns of validity and generality of the final well-being surveys. Additional validity studies with different samples of schools will be required to confirm the validity and generality of the teacher's well-being surveys.

The preliminary teachers' well-being surveys included IEQ-related items that were identified as potential manifestations of teachers' well-being in stage one of this study. For instance, the preliminary TPWB survey included the items "I am happy because I control air temperature in my classroom" and "The amount of daylight in my class affects my mood (−)". Additionally, the preliminary TSWB survey included the item "High background noise in class affects my interaction with students (−)". Results of the EFA showed that none of these IEQ-related items loaded significantly to the respective teacher's well-being survey, except for the more generic TPWB item "My classroom is not fit for teaching and learning" which doubled as the most significant item of the "Job satisfaction and professionalism" factor. Hence, the EFA results suggest that those IEQ-related items were insignificant manifestations of TPWB and TSWB respectively. Nevertheless, some IEQ-related items were significant manifestations of TPhWB. These included the items "I feel uncomfortable due to the level of air temperatures in my classroom (−)" and "I frequently raise my voice in class due to high background noise (−)" under the factor "Role and Work Environment Outcomes".

The two-tailed correlation tests between the refined IEQ satisfaction factors and the existing well-being surveys suggested that the former are probably insignificant measures of well-being except for $VTC_{sat}$ which significantly correlated with RAND 36. Given that the preamble to the IEQ survey items adapted for this study was satisfaction based just like other surveys in the IEQ literature, most of the IEQ factors were expected to correlate significantly with Ryff's scale of psychological well-being. However, the seemingly hedonic underpinning of IEQ satisfaction may explain why its factors did not significantly correlate with the eudaimonically inclined Ryff's scale. This is because psychological well-being measures are either hedonic or eudaimonic; with the former focusing on human happiness and satisfaction and the latter on "optimal functioning", and "self-actualization" [16, p. 661]. This reinforces the need for IEQ satisfaction questions that are eudaimonically inclined given that building indoor environments are expected to enhance human functioning in all domains of life. The correlation tests showed that $VTC_{sat}$ was significantly associated with the three final teachers' well-being measures while $L_{sat}$ and $AP_{sat}$ significantly correlated with the TPhWB measure only. These significant associations between the IEQ satisfaction factors and the final teachers' well-being measures can be attributed to the contextualization of these new teachers' well-being measures. Only $VTC_{sat}$ correlated significantly with both RAND 36 and the final TPhWB, thus, suggesting that $VTC_{sat}$ is a significant measure of TPhWB. In contrast, the lack of concurrent significant associations of $L_{sat}$ and $AP_{sat}$ with the final teacher's well-being surveys and their corresponding existing well-being surveys, suggest that both $L_{sat}$ and $AP_{sat}$ were less significant measures of well-being. This finding is in contrast with Aries et al. [39] who found a significant relationship between view quality and physical and psychological comfort. These insignificant correlations of $L_{sat}$ and $AP_{sat}$ with the well-being measures in this study are probably because most of the teachers were very satisfied with both IEQ factors and not because those factors are not measures of teachers' well-being. Additionally, IEQ satisfaction surveys, in general, appear to be deductive and significantly influenced by the need to correlate objective and subjective assessments. Probably, an inductively conceived questionnaire with assessment items grounded in the qualitative IEQ perceptions of teachers would have led to significant associations with the three contextualized final teachers' well-being measures developed in this study.

The tests of associations between the three final teachers' well-being measures and the refined IEQ satisfaction factors in this study suggested that adverse IEQ is likely to have a direct physical impact first, which would in turn indirectly impact psychological and social well-being, as depicted in Fig. 2. Solid lines (Fig. 2) indicate a direct impact, long dashes indicate a likely indirect second stage impact, whiles long dash dot indicate other potential direct impact of IEQ on well-being. Although $L_{sat}$ and $AP_{sat}$ did not significantly correlate with RAND 36, the direct impact on physical well-being is suggested because of their significant correlation with the final TPhWB measure. This is also the case for the suggested direct impact of $VTC_{sat}$ on psychological and social well-being due to its significant correlation with the final TPWB and TSWB measures. The design of this study did not allow for correlation tests between the three measures of teachers' well-being. However, an earlier study [50] concluded that physical well-being has a positive impact

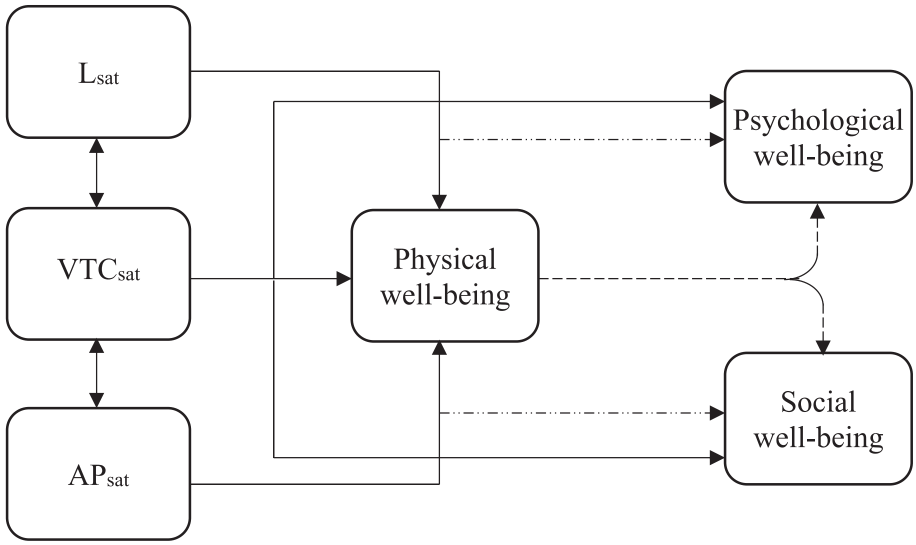


**Fig. 2.** Hypothetical two stage impact of IEQ satisfaction factors on occupants' well-being.

on psychological well-being. Additionally, it seems logical to expect physical well-being to have an impact on social well-being, hence, the predicted indirect impact of IEQ satisfaction factors on psychological and social well-being through physical well-being. These suggested impact relationships will require further research (i.e. path analysis or structural equation modeling) for verification and validation, ideally using inductively conceived measures of IEQ satisfaction and well-being.

## 6. Conclusions

This research investigated the use of occupants' IEQ satisfaction factors as measures of school teachers' well-being. The results showed that the newly developed TPWB, TSWB, and TPhWB surveys are valid measures of the corresponding subdivision of teacher's personal well-being. These developed and validated well-being surveys are conceptually related to their corresponding context-free well-being surveys; however, they are unique given that they also include factors that are relevant to teachers in schools and not other life domains. Hence, the new well-being surveys are likely to produce a more representative assessment of those subdivisions of teachers' well-being in school compared to their corresponding existing surveys. The $VTC_{sat}$ factor of IEQ satisfaction was the most significant measure of TPhWB, however, the moderately positive correlations suggest that TPhWB cannot be limited to $VTC_{sat}$. Additionally, the lack of concurrent associations of $L_{sat}$ and $AP_{sat}$ with the corresponding measures of the two sets of well-being surveys only implies that both IEQ factors were insignificant measures of well-being in this study and not that they are not measures of occupants' well-being.

Results of this study provide evidence which suggests that occupants' IEQ satisfaction should not generally be misconstrued to be equivalent to occupants' well-being. However, a normed practice of measuring both IEQ satisfaction factors and other subdivisions of occupant's well-being (teacher and building occupants in general) should probably be the preferred strategy moving forward. Considering the deductive nature of most existing IEQ satisfaction surveys, inductively conceived surveys may yield different results where most IEQ factors will align with occupants' well-being and lead to identifying significant perception patterns over time. These patterns will likely feed into IEQ enhancement strategies to maximize teachers' well-being in school, ultimately benefiting the teaching and learning process. Additionally, the hypothetical model shown in Fig. 2 could serve as the starting point for exploring the impact mechanism of IEQ satisfaction vis-à-vis teachers' well-being, and other building occupants.

The sample size in this study was found to be satisfactory for the EFAs that were conducted; however, the sample size is relatively small and thus minimizes the generality of the results. Further studies in different schools with larger samples of teachers will be required to validate the findings of this study. The $VTC_{sat}$ factor of IEQ satisfaction included the item “Your ability to … control physical conditions (such as temperature, light etc.) in your classroom”. The reference to control of lighting and other IEQ parameters can potentially lead to conflicting conclusions, hence, future studies should consider including different control items to relevant factors of IEQ satisfaction. Furthermore, future studies should consider conducting controlled experiments or employ other strategies of inquiry where participants will be exposed to varying levels of indoor lighting and acoustics parameters to assess the associations of $L_{sat}$ and $AP_{sat}$ with the subdivisions of occupants' well-being. The refined teachers' IEQ satisfaction survey and the final TPWB, TSWB, and TPhWB surveys will be used for further analysis towards meeting the goal of the broader research study at the University of Manitoba. However, IEQ researchers and practitioners are encouraged to use these measures to determine their confirmatory validity and required improvements.

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## Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.buildenv.2017.03.045.

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